Research Statement

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Overview: Measuring Representation in Collective Decisions

Many collective decisions are now made with the help of algorithms. Cities let residents vote on how to spend part of their budget, and a voting rule decides which projects are funded; parliaments are elected under different electoral systems; and large language models are being tested as tools to summarize public consultations and to draft statements that a whole group can agree on. In each case we can ask: *does the outcome represent the people it affects?* To answer this with evidence rather than intuition, we need two things: a way to compare decision problems (is this election similar to that one, and are synthetic test cases similar to real ones?) and real data to test against. Computational social choice, the field that studies voting rules with the tools of computer science, long evaluated rules mainly through proofs and through experiments on synthetic votes, often drawn uniformly at random, with no general method to check how close these votes were to real elections.

My work provides both:

- *Maps of elections.* I proposed the *positionwise distance*, which compares two elections by how often each candidate is ranked in each position, after optimally matching the candidates of one election to those of the other. It therefore does not depend on the names of the candidates or on the order of the voters. Computing this distance between all pairs in a collection of elections and drawing them in the plane gives a *map of elections*, on which similar elections lie close to each other. This makes it possible to see which models of synthetic votes produce elections that resemble real ones (AAMAS'20; AIJ'25).
- *Pabulib.* I initiated Pabulib, the largest open repository of participatory budgeting data. It holds over 2,000 elections from 8 countries and has been cited in more than 70 papers (IJCAI'23).
- *The Method of Equal Shares in practice.* Under this rule, each resident receives an equal share of the budget, and a project is funded only if its supporters can jointly pay for it from their shares, so that groups of residents with different priorities each get a fair part of the budget. I am a member of the team that convinced Polish cities to adopt it, and in 2025 I co-founded *Science for Democracy*, a nonprofit that works to bring the Method of Equal Shares to Polish municipalities.

As a PhD student, I was the principal investigator of a National Science Centre grant (NCN Preludium) on the *election isomorphism problem*, from which maps of elections grew. My PhD thesis, *Map of Elections*, supervised by Piotr Faliszewski, received the Polish Prime Minister's Award (2025). I also received a START fellowship of the Foundation for Polish Science (2023) and the AAMAS 2023 Best Student Paper Award. As a CNRS postdoc at LAMSADE (Paris Dauphine-PSL) with Jérôme Lang (2024–2026), I worked at the intersection of social choice and machine learning. At the University of Geneva, I build NLP pipelines that extract structured information from diplomatic and policy texts, and in a paper to appear at EMNLP 2026 we use LLM agents to simulate democratic deliberation and compare how well different electoral systems represent the variety of the agents' preferences (EMNLP'26).

At EPFL, I will extend these tools to collective decisions in which opinions are expressed in text and aggregated by AI, such as summaries of public consultations, consensus statements, and AI agents that vote on people's behalf. For example: given the comments that residents submitted in a consultation and an AI-written summary of them, how can we measure which groups of residents the summary leaves out?

Past and Current Research

How similar are two elections? A geometry of collective decisions. Experiments in social choice compare voting rules on collections of elections, real or synthetic. What such an experiment shows depends on which elections it contains: are the synthetic ones similar to real ones, and do they cover the situations a rule will face in practice? Answering this requires a distance between elections that ignores what the candidates are called and in which order the voters are listed, since two elections can have the same structure with entirely different candidates (presidential candidates vs. types of sushi). We showed that the most natural such distances, and the related problem of finding one election inside another, are computationally hard (AAAI'19; JCSS'25; AAAI'22; JAIR'24). The positionwise distance, described above, is my answer: it can be computed efficiently, and it made it possible to draw the *map of elections* (AAMAS'20), to which we then added landmark elections that make its regions interpretable (IJCAI'21). The complete framework appeared in *Artificial Intelligence* (AIJ'25).

The map turned a picture into a way of running experiments: placing the elections of an experiment on the map shows which parts of the space it covers, which it misses, and where real elections lie relative to synthetic ones. The same geometry applies well beyond ordinal elections. My coauthors and I built maps of approval elections (IJCAI'22), of elections of different sizes (AAAI'25), of stable roommates and stable matching instances (AAMAS'23 *Best Student Paper Award*; JAIR'24), of tournaments (ECAI'25), and of fair division instances (AAAI'26), and connected it to preference learning (NeurIPS'22). To help others use this methodology, I co-authored a guide to numerical experiments on elections (IJCAI'24) and released two Python libraries: `mapel`, for drawing maps, and `prefsampling`, for sampling synthetic preferences.

Discovering structure in preferences. Beyond comparing elections, I develop measures of what is *inside* them. Polarization has long been studied in political elections where voters name one party, but was almost entirely neglected for ordinal and approval ballots. We introduced indices of agreement, diversity, and polarization built on a single generalization of Kemeny's rule, *k-Kemeny*, which seeks k representative rankings instead of one consensus; the problem is NP-hard, and we gave efficient approximation algorithms and extended the indices to approval elections (IJCAI'23; IJCAI'26), then used them to quantify how diverse structured preference domains are (AAAI'26; AAMAS'26). During my postdoc with Jérôme Lang, I initiated and led a line of work that inverts the usual direction of the field: instead of designing a rule, we *discover hidden properties of a preference dataset*, such as clones, antagonistic candidates, and diverse groups of candidates (AAMAS'24; IJCAI'24), and, in ongoing work, hidden structure among voters, such as clusters of similar voters and outliers. This brings social choice close to the central task of machine learning: finding structure in data. Closer to political science, I designed voting rules that let allied candidates avoid the spoiler effect (AAMAS'24) and a nonparametric test for gerrymandering (IJCAI'24).

Real data and real deployments. In participatory budgeting (PB), residents vote on how to spend part of a city's budget. In 2020 I initiated *Pabulib*, with D. Stolicki and N. Talmon, and I have led its development since (IJCAI'23). It now holds 2,053 elections from 30 cities in 8 countries, with 8.6 million votes on over 38,000 projects; researchers abroad now persuade their own cities to contribute, which is how Amsterdam and Budapest entered it. Pabulib is the standard dataset for PB research and, increasingly, for approval voting in general: more than 70 papers have used it, including field studies of participatory processes in Switzerland and Taiwan. Real data opened questions that theory alone could not answer: learning realistic models of approval voting (AAMAS'25), measuring how well individual projects performed under a given rule (IJCAI'24), and a game-theoretic analysis of how proposers set project costs under the Method of Equal Shares (NeurIPS'25).

This work reaches practice. The analyses in the Pabulib paper were used by the team, of which

I am a member, that convinced cities in Poland and across Europe to adopt the Method of Equal Shares, starting with Wieliczka, the first city to use it. Proportional rules sometimes fund a costly project over a cheaper and more popular one, which is correct but surprising to residents, so I have also worked on ways of presenting outcomes that explain such decisions. In 2025 I co-founded *Nauka dla Demokracji* (Science for Democracy), a nonprofit that connects researchers with the officials who run democratic processes. Pabulib now operates under its auspices, closing the loop between the data that cities produce and the research that improves their processes.

Toward language. At CNRS LAMSADE, I designed simulation frameworks that model multi-agent decision-making, which collaborators at other institutions now use. At the University of Geneva (Global Studies Institute), I study collective decisions in international politics, combining voting records with NLP pipelines that extract structured information from large collections of diplomatic and policy texts. Two projects show this direction.

Simulating democratic deliberation with LLM agents. With collaborators at the University of Toronto and Google DeepMind, we built *VoteSim*, a framework that simulates a full democratic cycle with language-model agents (EMNLP'26). Voters are LLM agents conditioned on the demographic profiles and statements of real participants in a public dataset (PRISM). Parties draft policies on twelve divisive issues, voters rank the parties, seven electoral systems turn the same ballots into seven parliaments, and each parliament negotiates and passes a bill, which the voters then evaluate. Across all seven language models we tested, proportional systems produced bills that voters preferred to those of majoritarian systems, and the least satisfied fifth of voters gained the most. Because the voters are simulated, these results describe language-model agents, not real electorates; testing such findings against real ballots is part of Direction A below.

What an abstention means in the UN General Assembly. Abstention is the most common way states withhold support in the General Assembly: of roughly 860,000 votes cast between 1946 and 2025, 11.8% were abstentions and 5.5% votes against. Yet standard measures of state preferences place every abstention at the same point between support and opposition. In ongoing work, we classify all 101,070 abstentions by the context in which they were cast (alone, inside a large crowd of abstainers, or in a vote with an organized opposing camp), using only the votes of the other states and the abstaining state's voting bloc. Abstaining alone is concentrated among the permanent members of the Security Council, and the mix of contexts tracks international conflict even though the overall abstention rate does not: the share of abstentions cast in contested votes fell by more than half between the early Cold War and the 1990s and has risen since. To label the subject of each resolution, we embed its text with a language model and match it to the UN library's thesaurus of subjects.

Research Agenda

More and more opinions in collective decisions are expressed in text. Residents submit free-text proposals, consultations collect thousands of comments, AI systems are starting to summarize and mediate them, and AI agents that vote on people's behalf are being proposed. Recent work has made a promising start. *Generative social choice* (Fish et al., EC'24) gives proportional-representation guarantees for slates of LLM-generated statements, and the *Habermas Machine* (Tessler et al., *Science* 2024) showed that LLM-mediated deliberation can find common ground at scale. Three things are still missing: ways to *measure and audit* representation in text, an understanding of the *dynamics* that arise when AI agents take part repeatedly, and *validation on real data* rather than synthetic opinions. These are the problems that my work addressed for ballots. My agenda has three connected research directions; each poses concrete open problems and has a natural first PhD project.

Direction A: Representation when opinions are expressed in text and aggregated by AI.

- **Auditing representation in text.** Given individual opinions written in natural language and an aggregate text (a summary or a consensus statement), how should we measure how proportionally the aggregate represents the population? Which measures satisfy natural axioms, and which can be estimated with few LLM queries and provable error bounds? Existing methods *generate* representative text; we lack principled ways to *evaluate* it. My structure-discovery methods, which find clusters and outliers among voters, are the natural starting point for finding the groups that an aggregate text leaves out.
- **Polarization dynamics in AI societies.** Under which electoral systems, discussion structures, and mixes of model families does repeated deliberation among LLM agents converge, fragment, or polarize? I conjecture that majoritarian systems accelerate polarization and that diversity of model families slows it. My polarization indices provide the measures, and records of real multilateral negotiations provide a human baseline.
- **Faithful AI proxies, tested on real ballots.** Pabulib contains real approval ballots over projects with textual descriptions, which is rare ground truth for AI proxies. When an LLM votes on a person's behalf, how much does the *outcome* change under rules such as the Method of Equal Shares, and which rules are robust to proxy errors? This is the first PhD project of this direction.

Five-year goal: a theory of representation for text-based collective decisions, with audit tools that practitioners can apply to AI-generated summaries and consensus statements.

Direction B: Participatory democracy as a living laboratory.

- **Richer democratic data.** I will extend Pabulib with full ballots, project texts, residents' comments, and outcomes over several years. This will turn it into a longitudinal benchmark, including for Direction A.
- **Does the rule change the process?** Theory predicts that proportional rules change which projects are proposed and at what cost (NeurIPS'25). Field studies with partner cities can test whether they also change who participates and who proposes. This is the first PhD project of this direction.
- **Tools that cities and residents can trust.** Open tools to choose, simulate, and audit PB rules, and to *explain* their outcomes to residents, so that cities can adopt evidence-based designs without in-house experts and residents can see why a project was or was not funded.

Switzerland is an ideal setting for this work, with its tradition of direct democracy, Lausanne's participatory budget, and Swiss cities such as Aarau and Winterthur that have already used the Method of Equal Shares. I will partner with them as I do through Science for Democracy.

Direction C: A geometry of collective decision problems, from ballots to text.

- **Computing invariant distances at scale.** Which domains admit polynomial-time relaxations of isomorphism-invariant distances that, like the positionwise distance, preserve the geometry? Can sketching and sublinear techniques make them scale to elections with millions of voters? This is the first PhD project of this direction.
- **Benchmarks with provable coverage.** How many instances, and which ones, must a benchmark contain to cover the space of instances up to a given precision under a given distance? Which samplers achieve this? The answers will become defaults in `prefsampling` and `mapel`.
- **Maps of deliberation.** Extending the geometry from ballots to sets of textual opinions will make it possible to compare deliberations, human or AI, the way we now compare elections. This will give Direction A its evaluation backbone.

Students, funding, and service. The three directions offer PhD projects of different flavors: theoretical (C), empirical and NLP-oriented (A), and applied and data-centric (B). I have already co-supervised more than ten students, led a grant as principal investigator (NCN Preludium), and worked within an ERC-funded project. I will apply to the Swiss National Science Foundation early on and for an ERC Starting Grant built around Direction A. I serve on the editorial board

of JAIR (since 2026) and on the program committees of NeurIPS, AAAI, AAMAS, and IJCAI.

Why EPFL

My agenda sits between algorithms, AI, and the social sciences, and EPFL is one of the few places where all three are strong under one roof. I see concrete joint projects:

- with *Michael Kapralov* on sketching and sublinear algorithms for isomorphism-invariant distances between very large elections (Direction C);
- with *Robert West* on how faithfully LLMs represent the opinions of real populations, and on polarization dynamics in AI societies, combining his work in computational social science with my preference measures (Direction A);
- with *Antoine Bosselut* on measuring representation in LLM-generated summaries and consensus statements (Direction A);
- with *Bryan Ford* on secure, auditable platforms for participatory processes and digital deliberation (Direction B).

Beyond EPFL, the Lake Geneva region offers what this agenda needs most: direct democracy, municipalities that experiment with participation, and international organizations whose negotiations are rich records of real deliberation.